

Timing	Lesson description	Additional materials	Comments
JuniorKoders Python Lesson 13. GUI – Input & Output Part 2			
Lesson objectives: <i>Introduce students to GUI input and output of Python: Check Button, Combo Box, Text Area.</i>			
5	Last Lesson Review - Ask students about what they have learned during the last lesson. - Ask students if they still remember what they learnt last week.		
5	Introduction to Input & Output Elements in Python GUI Tell students they are going to continue learning about input and output in Python GUI. Ask if students know about some elements which can be used as input/output. Explain that some elements can be both input and output, even buttons. The teacher can show the main project, so the students have an idea of what they are going to learn today. Ask the students if they have studied HTML before because what we will be doing is similar to making an HTML Form. Summarise: the lesson today will be about input and output of Python GUI. It will be similar to creating a form because we have elements such as textbox, radio buttons, checkboxes, and many more.	5. Project Lesson 13.py	

10	Create Check Button Widget Show & Explain: Check button widget ● Tell students another commonly used widget is the checkbox. Some people also call them the check button widget when using Tkinter for Python. ● Show the project to the students to create checkboxes, and it is similar to creating a radio button. Although, for checkboxes, we do not need to include the “value” option. Apply & Code: Create Checkbox ● In the sample project, we have four examples of checkboxes. To create the checkboxes, we would need a unique variable for each one. ● By default, the checkbox will be unchecked. To change the state of it, we can use different commands such as <code>select()</code>, <code>deselect()</code>, or <code>toggle()</code>. ● To group any checkboxes, we would need to create a new variable to be set the group, similar to when creating radio buttons. ● If we set the variable to true, the checkbox will be automatically checked in the beginning. ● Make sure we use the Boolean variable if we want to make true or false ● There are two new options which we can use for checkboxes, “onvalue” and “offvalue”. This is to set the value of the checkbox when it is checked, or unchecked. We can later print out the value to see in the shell.	1. Create Checkbox.py	
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	Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname		
15	Create a Combo Box Widget Show & Explain: Combo Box Widget ● Tell students that a combo box is a combination of the entry widget and drop-down menu. We need to import another module of Tkinter called ttk. Apply & Code: Create Combo Box ● First, we need to create a variable to store the combo box. Then to create the combo box, the code is slightly different because of the ttk module: ○ <code>tkinter.ttk.Combobox()</code> ● After creating the combo box, we can specify the contents of it: ○ <code>combo["values"] = ("item_1", "item_2", "item_3", "item_4")</code> ● Then we can set the current value for the combo box, the value which will be seen by default. ● We can also add new values to the combo box. ● To connect command to the combo box, we need to first use the “bind()” method. ● “<<ComboboxSelected>>” is a virtual event which occurs when the user selects an element from the list of value. ● We can include a function without defining it using the “lambda” function. ● In our example, we created a function to focus on the label, instead of the combo box, so it looks neater because the combo box will not be highlighted. This is so the appearance of the combo box is not too distracting.	2. Create Combo Box.py	

	• Lastly, there are different states which we can apply to the combo box. In our example, we have the “readonly” and “disabled” state. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname		
10	Create Lambda Expression Show & Explain: Lambda Expression • Tell students the lambda function is usually used to create a function within another defined function. Lambda is used because Tkinter will not accept a function with any arguments. To bypass such issue, we will be learning how to use lambda. • Show the project to the students to let them see the lambda function can take any number of arguments, but only one expression. Apply & Code: Lambda Example • First, create a variable to store the function. • Then type the “lambda” function. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	3. Lambda Example.py	
10	Create a Text Area Show & Explain: Text Area • Tell students a text widget is used for creating a multi-line text area. It can be used for a wide range of tasks. One of the primary purposes is to provide simple multi-line areas, as they are often used in forms. We create a text widget by using the Text() method.	4. Create a Text Area	

- **Show the project to the students** to let them see the similarities and differences between an entry widget and a text widget. It is like comparing a textbox and a text area in HTML.

Apply & Code: Text Area

- First, we create a text area similarly to creating an entry widget.
- We can add the height and width to our text widget. If we do not specify any dimensions, the text area will become too large.
- We can insert text into the text widget from our coding using the **insert()** function.
- We can also get the text inside our text area similarly to our entry widget. We do it by defining a function.
- The difference from what we did to our entry widget is the attributes inside the **get()** function. Our get function needs to include the **"1.0"** this is to read the input from line 1 and starts from character 0. As for the **"end-1c"**, this is to get the last character of the input. If we only put "end", we will get an extra new line which we do not need. To erase the new line, we use **"end-1c"**, which means it will erase one character, thus erasing the extra newline.
- After creating the text area, we can create a button, which, when clicked, will run our function. We also create an empty label, where we can print out the content of the text area.
- Lastly, we just place the elements according to where we want them to be.

Save the project – ask students to save the project in a specific place with a proper name:
 StudentName_Projectname

30	Mini project Show & Explain: Project Lesson 13 • Tell students that they are going to create a mini-project by themselves. • Show the project to the students and remind them of the different things they have learnt in the lesson. • Explain that the project is the combination of what they learnt last week and today. Apply & Code: Project Lesson 13 • First, the students need to create two entry widgets for entering their names. • Then there is the spin box for selecting age. • They can also choose their gender using radio buttons. • There is a combo box to choose some texts. • Checkboxes are also present in the project. • Lastly, they can press the button to print out a text combining everything they have input. • We can create the text area as a comment box, so the users can type in their comments, and post it in the label of the window. Save the project – ask students to save the project in a specific place with a proper name: StudentName_Projectname	5. Project Lesson 13.py	
5	Last Overview Summing Up • We sum up by asking questions making sure the students remember the most important things. • We thank students for doing a great job overall.		

